MCQ v3 load test: can 35 students sit one test at once?

quiz.batchllm-workspace.info · build 5227596 · 17 Oct 2026 · simulated students on a 30-question paper with an 8 KB context block

Yes. 35 at once ran clean, with room to spare.

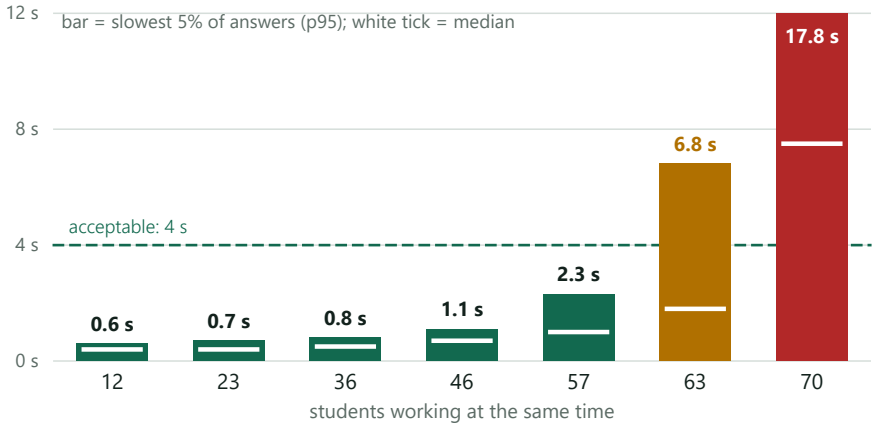
35 students signed in, started and **all submitted within one minute** with zero errors and every answer stored exactly as clicked. The site stays fast up to **~50 students working at once** and falls over around **65**.

35 STUDENTS, EVERYTHING AT ONCE (HARDER THAN A REAL CLASS)

Action, 35 at once	Median	Slowest 5%	Worst
Sign in (all within 10 s)	**1.9 s**	2.7 s	2.9 s
Start quiz (all within 5 s)	**6.3 s**	7.2 s	7.3 s
Save an answer (5–15 s pace)	**0.64 s**	1.9 s	3.3 s
Submit (all within 56 s)	**1.65 s**	2.5 s	3.1 s

- **0 errors, 0 lost answers.** Every stored answer and score was checked against what each student clicked.
- Start is the slowest moment: 35 shuffled papers are built and drawn in the same 5 s.
- Server health never went above 0.8 s.

HOW MANY AT ONCE? REALISTIC PACE (ONE ANSWER EVERY 20–40 S)



Reading the chart

- **Up to ~46:** under 1.1 s even for the slowest answers, the same as one student alone.
- **~57:** still fine (slowest 2.3 s).
- **63:** slipping (6.8 s). **70+:** a cliff where everyone stalls; the test stopped itself.
- So **35 has ~15–20 students of headroom**.

HOW WE GOT HERE

Original build (Community Cloud)

- 35 students: app **stalled 7 minutes**, 17 disconnected, only 15 submitted.
- **17 answers shown as picked but never saved.**
- Cause: one shared Supabase connection corrupted by simultaneous clicks.

Fix 1: e3f7fe4 (Community Cloud)

- No more errors or lost answers.
- Only the clock on a timer; shared quiz data read once.
- But still **capped at ~20** by the server's small CPU share.

Fix 2: 5227596 + new host

- Each answer redraws **one question, not 30**.
- Countdown ticks in the browser; server checks every 15 s.
- **35 clean; ceiling ~50–65.**

BEFORE THE REAL CLASS

1. **Avoid everyone pressing Start in the same second** if you can; that is the one ~6 s wait.
2. Roughly 1 “Saved” tick in 300–1000 answers can be missing (a Streamlit timing quirk). The answer is stored and the progress count corrects within 15 s, so students needn't re-click.
3. Re-run the 35-student test after any change to the quiz-taking page or the hosting (loadtest/REPORT.md has the commands).

Method. Each simulated student is a real Streamlit session over the app's own websocket protocol (loadtest/swarm.py): sign in, Start, answer 30 shuffled questions, then submit or let time run out. Latency = click to updated screen (browser drawing time excluded). A real student answers about once a minute, so the 5–15 s pace above is a stress test; the chart uses 20–40 s. The live site's health was watched throughout and the ramp aborted itself at the cliff. All ZZTEST test data was deleted afterwards. Full results, including the original Community Cloud runs: loadtest/REPORT.md.